

Reflection Paper — Wholesale Customer Segmentation

What did you implement?

In this project, I implemented an unsupervised learning approach to segment wholesale customers based on their purchasing behavior. The main goal was to group customers with similar buying patterns without using predefined labels.

I used six spending-related features from the dataset:

- Fresh
- Milk
- Grocery
- Frozen
- Detergents_Paper
- Delicassen

First, I applied **K-Means Clustering** to identify customer groups based on similarities in their spending patterns. The algorithm grouped wholesale customers into different clusters according to their purchasing behavior.

I also implemented **Agglomerative Hierarchical Clustering** as an additional clustering algorithm. This allowed me to compare two different unsupervised learning approaches and understand how each algorithm segments customers.

Segment Interpretation

Cluster 0 — High Fresh and Grocery Spending Customers

This cluster contains customers who spend more on products such as Fresh, Milk, and Grocery items. These customers appear to be regular buyers with high demand for daily-use products.

Business Action:

The distributor should provide bulk discounts, loyalty programs, and special offers to maintain these valuable customers and encourage larger purchases.

Cluster 1 — Lower Spending Customers

This cluster represents customers with lower spending amounts across most product categories. They may represent smaller businesses or customers who purchase less frequently.

Business Action:

The distributor can provide promotional offers, smaller packages, and targeted marketing campaigns to increase their purchasing activity.

Cluster 2 — Specialized Product Buyers

This cluster includes customers who have higher demand for specific categories such as Frozen products, Detergents_Paper, or Delicassen compared with other categories.

Business Action:

The distributor should create customized product bundles and recommend products based on the customer's purchasing behavior.

Understanding K-Means

K-Means is an unsupervised machine learning algorithm used to divide data into groups called clusters. The algorithm requires selecting a value called **K**, which represents the number of clusters.

K-Means works through an assign-and-update process:

1. The algorithm selects K initial centroids (cluster centers).
2. Each data point is assigned to the nearest centroid based on distance.
3. The centroids are updated by calculating the average position of points in each cluster.
4. These steps are repeated until the clusters become stable.

The final result is a set of groups where customers inside the same cluster have similar purchasing behaviors.

Second Algorithm

For the second algorithm, I selected **Agglomerative Hierarchical Clustering**.

I chose this algorithm because it provides a different clustering approach compared with K-Means. Instead of using centroids, Agglomerative Clustering creates clusters by gradually merging similar data points together.

From my research, I learned that Agglomerative Clustering starts by treating each data point as an individual cluster. It then combines the closest clusters step-by-step until the required number of clusters is reached.

Advantage:

One advantage of Agglomerative Clustering is that it can identify complex relationships between data points and does not depend on random centroid initialization.

Limitation:

One limitation is that it can require more computational resources when working with large datasets.

The **Silhouette Score** was used to compare the performance of K-Means and Agglomerative Clustering. The algorithm with the higher Silhouette Score provides better-separated and more meaningful clusters.

Findings

After comparing both clustering approaches, I would recommend **K-Means Clustering** for this wholesale customer segmentation task. K-Means is simple, efficient, and works well with numerical spending data. It also produces clusters that are easier for businesses to understand and use for decision-making.

Agglomerative Clustering is also useful because it provides another view of customer relationships and can discover different grouping patterns. However, for a wholesale distributor that needs a practical and scalable solution, K-Means is the better choice.

Overall, clustering helped identify different customer groups based on their purchasing behavior. These insights can help the distributor improve marketing strategies, create personalized offers, and build stronger customer relationships.